INDIVIDUAL PROJECT REPORT

Review 1 — Problem Identification & Proposal (24/8/2026)

AI-Driven Edge–Cloud Network for Cyclone Prediction and Resilient Ocean Telemetry

Student	Register No.	Team No.	My role
Laksh Ahuja	24BDS0212	_____	Central server, dashboard, network QoS, cyclone simulation and testing

Team: Laksh Ahuja (24BDS0212), Anvay Sharma (24BCE0824), Sreehari (24BDS0417). **Repository:** https://github.com/laksh-ahuja06/AI-DRIVEN-EDGE-CLOUD-NETWORK

Summary

This report states the problem our team is solving, the evidence that it is real, what earlier work has tried, and the solution we propose. My own focus is the back half of the system: how cyclone telemetry should be prioritised on a slow link, and how a central server should turn incomplete data into a reliable cyclone-risk picture on a live dashboard.

1. Problem Statement

During a cyclone, ocean buoys and coastal stations must push their readings over slow, unreliable links, so the most important readings arrive late or not at all — exactly when forecasters and coastal residents need them most.

- **Why it matters:** cyclone warnings save lives only if they are timely; stale or missing pressure and wind data delays evacuation decisions.
- **Who is affected:** coastal and island communities and fishers on India's east coast, and the district disaster-management officials who order evacuations.
- **If unsolved:** a monitoring network treats a routine battery report and a falling-pressure alarm equally. One 2.1 MB satellite frame occupies a 9.6 kbps backhaul for about **29 minutes**; any alarm queued behind it waits that long, and gaps from packet loss leave the dashboard blank or wrong during the critical hours.

Objectives of My Part

Objective	Measurable success criterion
Server keeps a per-node history and detects every missing sequence number	100 % of gaps detected in test runs
Fused cyclone-risk output and alert available through an API	/predict answers in < 1 s; alert at risk ≥ 60
Critical readings forwarded ahead of routine traffic on a slow link	Critical latency well below normal traffic under load
Live dashboard that shows estimated vs measured data	Updates at least every 2 s
Network topology emulated with routing	End-to-end ping buoy → server succeeds

2. Evidence the Problem is Real

Table 1: Evidence that the problem is real

Source	Published fact / figure	Why it matters for my part
India Meteorological Department [1]	India has a coastline of about 7,500 km; the 1999 Odisha super cyclone killed about 10,000 people.	The people at risk depend on coastal data reaching the warning centre on time.
Esposito et al., 2022 [2]	Review of IoT early-warning systems: finds gaps in the literature and recommends adding a fog/edge layer to warning architectures.	Supports making decisions near the data while a central server keeps the overall picture.
Engineering estimate (our calculation)	4 bands × 512 × 512 pixels × 2 bytes = 2,097,152 bytes; at 9,600 bit/s: 2,097,152 × 8 / 9,600 ≈ 1,748 s ≈ 29 min.	Shows why the link must decide what to send first — the core of my QoS work.

3. Literature Survey

Table 2: Literature survey (15 articles, 2018–2022)

#	Author / Year	Method used	Result	Limitation
1	Yu et al., 2018 [3]	Survey of edge computing for IoT	Edge cuts latency and backhaul load	No priority by event severity
2	Abbas et al., 2018 [4]	Mobile edge computing survey	Architectures and offloading	Assumes reliable cellular links
3	Hassan et al., 2018 [5]	Taxonomy of edge paradigms for IoT	Requirements and open issues	No evaluation under loss

#	Author / Year	Method used	Result	Limitation
4	Pham et al., 2020 [6]	MEC in 5G and beyond survey	Resource allocation methods	5G setting, not marine links
5	Zhou et al., 2019 [7]	Edge intelligence overview	AI split across device, edge, cloud	Generic; no disaster case
6	Esposito et al., 2022 [2]	Review of IoT early-warning systems	Edge/fog layer recommended	Floods, quakes; not cyclones
7	Xu et al., 2019 [8]	Review of marine IoT monitoring	Catalogues ocean sensing systems	Little on congestion or gaps
8	Centenaro et al., 2021 [9]	Satellite IoT survey	LEO links for remote sensors	Low data rates; no scheduling
9	Kodheli et al., 2021 [10]	Survey of satellite communications	Backhaul options and limits	Not application-aware
10	Bennis et al., 2018 [11]	URLLC: tail latency and risk	Design for worst case, not average	Theory; 5G radio focus
11	Popovski et al., 2018 [12]	Building blocks for URLLC	Reserve resources for urgent data	Needs 5G infrastructure
12	Yates et al., 2021 [13]	Age of information survey	Freshness metric for status updates	Not tied to risk level
13	Fuller et al., 2020 [14]	Digital twin review	Live model mirrored from sensors	Assumes complete data
14	Chen et al., 2020 [15]	Review of ML for cyclone forecasting	Surveys ML forecast models	Assumes data already arrived
15	Chavas & Knaff, 2022 [16]	Simple model of storm wind radius	Realistic wind structure in < 1 s	Model only, no network

Gap: edge and MEC surveys move computing closer to the data but do not rank traffic by event severity; URLLC and age-of-information work shows how to protect urgent, fresh data, but on 5G infrastructure that coastal buoys do not have; and cyclone ML work assumes the data has already arrived. No work we found ties link priority to the *live* cyclone risk computed at the server, while also repairing missing readings and showing what was estimated. My part targets exactly this link between risk and the network.

Foundational methods we also build on (not counted in the 15): the Holland storm profile [17] and DiffServ class-based forwarding [18].

4. Proposed Solution Form

Form: coding and modelling. We will build the system as software — simulated sensor nodes, an edge gateway and a central server — plus a network topology in Cisco Packet Tracer. Fabrication or a product first would be wrong for this problem: we cannot deploy buoys at sea, and the hard part is the *behaviour* of the network under congestion and loss, which software lets us measure and repeat. Low-cost hardware nodes can be added later without changing the software design.

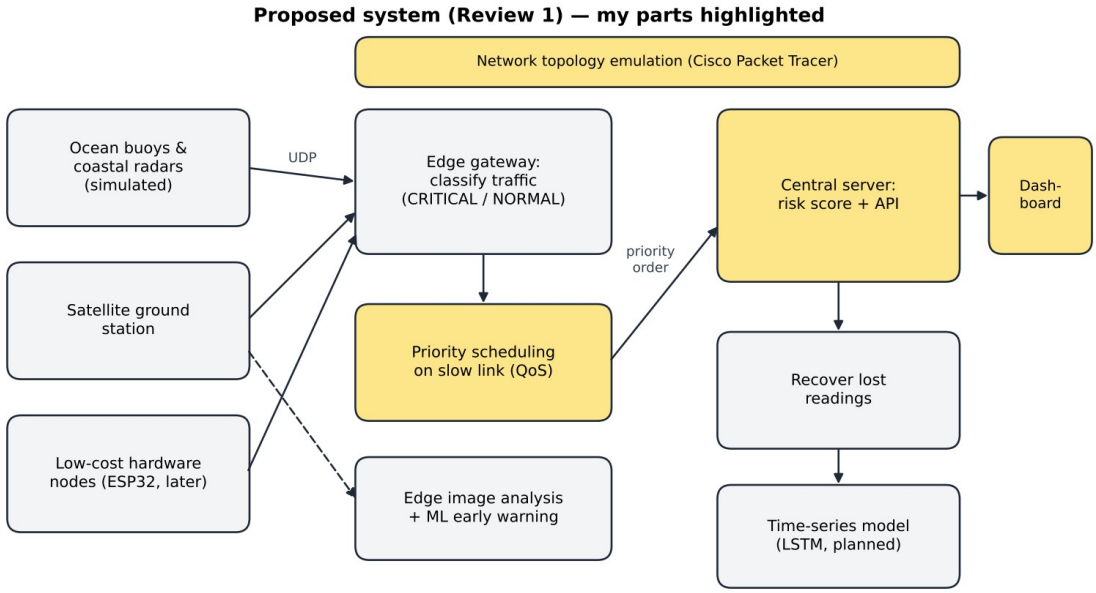


Figure 1: Proposed block diagram; my parts (server, dashboard, QoS, network emulation) are highlighted

Planned Method for My Part

Component	Planned method
Central server	Python FastAPI service with a background UDP listener; per-node ring buffer of recent readings; sequence-number tracking to detect gaps; risk score from wind and pressure; REST endpoints.
Dashboard	Single HTML page that polls the API and highlights CRITICAL nodes and estimated values.
QoS on the slow link	Start with a priority queue (critical first); later model the link capacity explicitly so that urgent data never waits behind routine data.
Simulation and testing	A cyclone scenario for the sensors, Packet Tracer for routing, and scripted end-to-end runs so every result can be repeated.

5. Initial Technical Feasibility

Table 3: Feasibility of my part

Area	Status
Skills I have	Python, sockets and basic networking, HTML/JavaScript, Git
Skills I must learn	FastAPI web framework, queueing / QoS theory (token bucket), Cisco Packet Tracer routing (OSPF), writing automated tests with pytest
Tools needed / access	Python 3, FastAPI, uvicorn (free); Cisco Packet Tracer (free with NetAcad account); own laptop — all available
Risks	No real buoy data (mitigation: physics-based simulation); timing bugs between threads; scope creep if every feature is attempted before Review 2

Table 4: Risk register for my part

Risk	Likelihood	Impact	Mitigation
Thread timing bugs between listener and API	Medium	High	Start threads in framework hooks; test end to end
No real buoy data	High	Medium	Physics-based simulation; real archives later
Packet Tracer cannot run our Python code	High	Low	Use it for routing; run code on loopback / Mininet
Scope creep before Review 2	Medium	Medium	Weekly milestones; baseline first

6. Team Role Split and Timeline to Review 2

Table 5: Team role split and interfaces

Member	Role	How their work meets mine
Laksh Ahuja (24BDS0212) — me	Central server, dashboard & network QoS	I own everything after the gateway: link scheduling, the prediction server and its API, the dashboard, the cyclone scenario used in demos, and the automated tests
Anvay Sharma (24BCE0824)	Edge gateway & edge intelligence	His gateway labels each packet; my scheduler and server trust that label, so we agree the class names and thresholds together
Sreehari (24BDS0417)	Sensing layer, hardware & data recovery	His nodes produce the sequence numbers my server checks, and his gap-filling function runs inside my server

My role is distinct: everything from the moment a packet leaves the gateway — scheduling, the server, the dashboard — plus the simulation and tests that prove it works.

Table 6: My weekly plan from Review 1 to Review 2

Week	Dates	My milestone	Deliverable / check
1	25–31 Aug	Agree the telemetry record format with the team; build the Packet Tracer topology (routers, subnets, OSPF)	Ping from a buoy PC to the prediction server succeeds
2	1–7 Sep	Server skeleton: UDP listener, per-node storage, risk score	Server receives packets from the gateway
3	8–14 Sep	REST endpoints and first dashboard; end-to-end test with teammates' modules	Working prototype demonstrated at Review 2

R1 Video Plan

Segment	What I show
Problem and evidence (1.5 min)	Face + screen; the IMD coastline / Odisha figures and my 2.1 MB-frame calculation
Literature gap (1.5 min)	Face + screen; my 15-paper table and the gap between network priority and cyclone risk
Proposed solution and feasibility (2 min)	Face + screen; the block diagram with my server, dashboard and QoS blocks
My role and timeline (1 min)	Face + screen; my weekly plan

References

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